

Assignment 3 — OOPs in Python

Q1. Student Class

```
class Student:
    def __init__(self, name, age, course):
        self.name = name
        self.age = age
        self.course = course

s1 = Student("Rahul Sharma", 20, "Data Science with AIML")
print("Name   :", s1.name)
print("Age    :", s1.age)
print("Course :", s1.course)
```

Output:

```
Name   : Rahul Sharma
Age    : 20
Course : Data Science with AIML
```

Q2. Car Class

```
class Car:
    def __init__(self, brand, model, price):
        self.brand = brand
        self.model = model
        self.price = price

    def display(self):
        print(f"Brand: {self.brand} | Model: {self.model} | Price: Rs.{self.price:,}")

car1 = Car("Toyota", "Innova", 1800000)
car2 = Car("Hyundai", "Creta", 1200000)
car1.display()
car2.display()
```

Output:

```
Brand: Toyota | Model: Innova | Price: Rs.18,00,000
Brand: Hyundai | Model: Creta | Price: Rs.12,00,000
```

Q3. Employee Class with Constructor (__init__)

```
class Employee:
    def __init__(self, employee_id, name, salary):
        self.employee_id = employee_id
        self.name = name
        self.salary = salary

    def display(self):
        print(f"Employee ID : {self.employee_id}")
        print(f"Name       : {self.name}")
        print(f"Salary      : Rs.{self.salary:,}")

emp = Employee(101, "Priya Mehta", 75000)
emp.display()
```

Output:

```
Employee ID : 101
Name       : Priya Mehta
```

Salary : Rs.75,000

Q4. Rectangle Class — Area Method

```
class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        print(f"Area of Rectangle = {self.length} x {self.width} = {self.length * self.width} sq. units")

rect = Rectangle(10, 5)
rect.area()
```

Output:

Area of Rectangle = 10 x 5 = 50 sq. units

Q5. Circle Class — Area Method

```
import math

class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        result = math.pi * self.radius ** 2
        print(f"Area of Circle (radius={self.radius}) = {result:.2f} sq. units")

c = Circle(7)
c.area()
```

Output:

Area of Circle (radius=7) = 153.94 sq. units

Q6. BankAccount — deposit() and withdraw()

```
class BankAccount:
    def __init__(self, account_holder, balance):
        self.account_holder = account_holder
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
        print(f" Deposited Rs.{amount:,} --> Balance: Rs.{self.balance:,}")

    def withdraw(self, amount):
        if amount > self.balance:
            print(" Insufficient funds!")
        else:
            self.balance -= amount
            print(f" Withdrew Rs.{amount:,} --> Balance: Rs.{self.balance:,}")

acc = BankAccount("Ananya Singh", 10000)
print(f"Account Holder : {acc.account_holder} | Opening Balance: Rs.{acc.balance:,}")
acc.deposit(5000)
acc.withdraw(3000)
acc.withdraw(15000)
```

Output:

```
Account Holder : Ananya Singh | Opening Balance: Rs.10,000
Deposited Rs.5,000 --> Balance: Rs.15,000
Withdrew Rs.3,000 --> Balance: Rs.12,000
Insufficient funds!
```

Q7. Animal & Dog — Method Overriding (Polymorphism)

```
class Animal:
    def sound(self):
        print("Some generic animal sound")

class Dog(Animal):
    def sound(self):          # Overrides parent method
        print("Bark")

a = Animal()
a.sound()

d = Dog()
d.sound()
```

Output:

```
Some generic animal sound
Bark
```

Q8. Person → Student — Inheritance

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

class Student(Person):
    def __init__(self, name, age, roll_number):
        super().__init__(name, age) # Call parent constructor
        self.roll_number = roll_number

    def display(self):
        print(f"Name      : {self.name}")
        print(f"Age       : {self.age}")
        print(f"Roll Number : {self.roll_number}")

stu = Student("Arjun Verma", 19, "BIT2024CS042")
stu.display()
```

Output:

```
Name      : Arjun Verma
Age       : 19
Roll Number : BIT2024CS042
```

Q9. Calculator Class

```
class Calculator:
    def add(self, a, b):
        print(f"{a} + {b} = {a + b}")

    def subtract(self, a, b):
        print(f"{a} - {b} = {a - b}")
```

```

def multiply(self, a, b):
    print(f"{a} x {b} = {a * b}")

def divide(self, a, b):
    if b == 0:
        print("Error: Division by zero!")
    else:
        print(f"{a} / {b} = {a / b:.2f}")

calc = Calculator()
a, b = 20, 4
calc.add(a, b)
calc.subtract(a, b)
calc.multiply(a, b)
calc.divide(a, b)

```

Output:

```

20 + 4 = 24
20 - 4 = 16
20 x 4 = 80
20 / 4 = 5.00

```

Q10. LibraryBook Class

```

class LibraryBook:
    def __init__(self, book_name, author, price):
        self.book_name = book_name
        self.author = author
        self.price = price

    def display_book_info(self):
        print(f" Book : {self.book_name}")
        print(f" Author: {self.author}")
        print(f" Price : Rs.{self.price}")
        print("-" * 35)

b1 = LibraryBook("Python Crash Course", "Eric Matthes", 599)
b2 = LibraryBook("Hands-On ML", "Aurelien Geron", 899)
b3 = LibraryBook("Deep Learning", "Ian Goodfellow", 1199)

for book in [b1, b2, b3]:
    book.display_book_info()

```

Output:

```

Book : Python Crash Course
Author: Eric Matthes
Price : Rs.599
-----
Book : Hands-On ML
Author: Aurelien Geron
Price : Rs.899
-----
Book : Deep Learning
Author: Ian Goodfellow
Price : Rs.1199
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```